Project Based Learning Report

On

Data Driven Analysis and Usage of Machine Learning in Precision Farming

By

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*In partial fulfillment of requirements for the award of degree in Bachelor of Technology in A*rtificial Intelligence and Data Science
(2027)

Under the Project Guidance of

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PROJECT REVIEW CERTIFICATE

This is to certify that the work recorded in this project report entitled **"Data Driven Analysis and Usage of Machine Learning in Precision Farming"** has been carried out by **Ms. Bidisha Bohidar & Ms. Suhani Terway** of Artificial Intelligence and Data Science Department of Sikkim Manipal Institute of Technology in partial fulfilment of the requirements for the award of Bachelor of Technology in Artificial Intelligence and Data Science. This report has been duly reviewed by the undersigned and is hereby recommended for final submission for Project Based Learning.

Date: 19/11/2024 Place: SMIT, Sikkim	**Supervisor Signature** Mr. Sital Sharma Asst. Professor Department of AI&DS Sikkim Manipal Institute of Technology; Majitar, Sikkim – 737136

DECLARATION

We, the undersigned, hereby declare that the report titled **"Data Driven Analysis and Usage of Machine Learning in Precision Farming"** is an original work submitted in fulfilment of the requirements for **Project Based Learning** at **Sikkim Manipal Institute of Technology (SMIT)**. This report has been prepared under the guidance of **Mr. Sital Sharma**, SMIT.

We affirm that this report is the result of our collaborative efforts, except where specific references are made and duly acknowledged. We further certify that all sources of information used in this report have been properly cited and referenced according to the appropriate standards.

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CERTIFICATE OF ACCEPTANCE

This is to certify that the below mentioned students of Artificial Intelligence & Data Science Department of Sikkim Manipal Institute of Technology (SMIT) are working under the supervision of **Sital Sharma** of SMIT on the project entitled **“Data Driven Analysis and Usage of Machine Learning in Precision Farming”**

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Lastly, we would like to extend our sincere thanks to all the staff members of Artificial Intelligence and Data Science department at Sikkim Manipal Institute of Technology. Their continuous support and guidance were instrumental in bringing our project to fruition.

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LIST OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABSTRACT

Precision Farming, an advanced agricultural approach that leverages technology and data driven techniques to optimize crop yields, proper utilization of all the resources, and environmental sustainability. This study explores the application of machine learning in precision farming, focusing on predictive analytics, real-time decision-making providing the best suggestion for the plantation of crop according to your suitable and favorable conditions by making it most feasible, viable and profitable at the same time.

This study emphasizes around collection and analysis of diverse datasets, including weather patterns, soil health, moisture content of soil, various crop growth stages, various season and their effects on various crops. Based on these parameters, the objective is to provide most suitable outcome with data backed recommendations to increase productivity while minimizing environmental impact.

1. Introduction:

Agriculture and allied activities in India engage nearly 45% of the country's population and contribute 19.9% to the gross domestic product (GDP) [1][8].

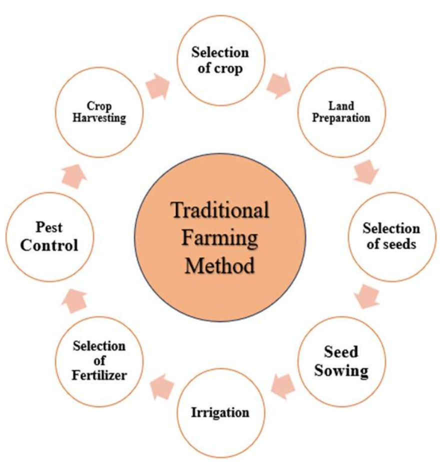
In developing countries like India, where agriculture remains the backbone of the economy, traditional farming methods face numerous challenges, including over-reliance on natural resources, unpredictable weather patterns, and inefficient use of inputs such as water, fertilizers, and pesticides. These issues often lead to low crop productivity and environmental degradation [5][4][10].

To address these challenges, precision farming practices powered by data-driven analysis and machine learning-based crop and fertilizer recommendation systems offer a transformative solution [3][9][11]. Precision farming uses advanced technologies like sensors, satellite imagery, and Internet of Things (IoT) devices to gather real-time data on soil health, crop conditions, and weather.

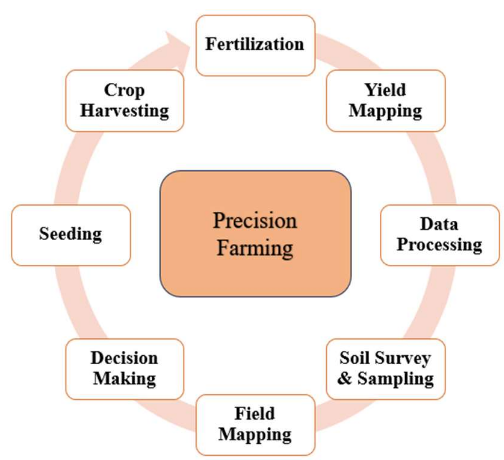
By analyzing this data with machine learning models, farmers can receive accurate, personalized recommendations for crop selection, fertilizer application, and irrigation schedules. This enables the efficient use of resources, reduces input costs, and increases crop yields [3][4]. The adoption of precision farming can revolutionize India's agricultural landscape, fostering economic growth and food security while promoting long-term environmental sustainability.

In countries like India, where smallholder farmers dominate agriculture, these AI-powered solutions can:

- Maximize land productivity
- Ensure sustainable farming practices
- Reduce the environmental impact of agriculture



[Fig 1. Traditional Farming Methods]



[Fig 2. Steps For Precision Farming]

2. Literature Survey:

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
01.	Soussi, A., Zero, E., Sacile, R., Trinchero, D., & Fossa, M. (2024). Sensors, 24(8), 2647	Smart Sensors and Smart Data for Precision Agriculture: A Review	The study presents a comprehensive review of **smart sensors and data integration** in precision agriculture, emphasizing technologies like IoT, AI, and big data. It discusses their role in optimizing crop management, improving sustainability, and resource efficiency. The paper also highlights global trends, challenges, and emerging innovations in smart agriculture technologies	-Limited exploration of the challenges posed by high sensor deployment costs and connectivity issues in underdeveloped regions. -Lack of detailed case studies focusing on diverse crop types and small-scale farms in developing nations	Supports the development of IoT-based, localized, cost-efficient solutions to aid resource optimization and sustainability in farming practices
02.	Reddy, G. V., Reddy, M. V. K., Spandana, K., Subbarayudu, Y., Albawi, A., Chandrashekar, R., ... & Praveen, P. (2024). In *E3S* Web of Conferences (Vol. 507, p. 01078). EDP Sciences.	Precision farming practices with data-driven analysis and machine learning-based crop and fertiliser recommendation system	This research paper tells us the importance of **soil conditions**, highlighting that urbanization leads to the degradation of soil quality, highly affecting the crop yields. Therefore, this paper introduces a new farming method called "**Precision Agriculture**" which makes the use of machine learning (ML) techniques and Internet of Things (IoT) devices to analyze soil data and recommend correct fertilizers suitable for the type of soil for the best crop yields This study also finds out that a specific ML technique, **Support Vector Machine (SVM),** helps farmers to make the best decisions for their crops.	There are a lot of research gaps in this study like limited awareness and understanding *of ML models, basic infrastructure* required to successfully implement these models, *data availability* (whether the farmers are well familiar with their land and soil), *privacy concerns* as the data provided can be invasive, and *capital* required, as investing in technological equipment can be costly.	It suggests how can we use ML in farming to optimize the use of resources to enhance the efficiency of crop growth in Agro dependent countries like India

[Table 1.1: Literature Survey]

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
03.	Nedoumaran, B., Prosper, A. M., Maddu, S., & Mithun, P. (2023, December). Precision Farming Using Machine Learning and Data Analytics. In *2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)* (pp. 1-7). IEEE.	Precision Farming Using Machine Learning and Data Analytics	Explores how machine learning and data analytics can revolutionize precision farming practices. By analyzing vast amounts of agricultural data, the paper demonstrates predictive models for soil moisture levels, crop health, and yield forecasting. The application of these techniques aims to optimize farming decisions, improving resource efficiency and enhancing productivity. Real-time data processing and insights facilitate smarter, data-driven agricultural practices.	Insufficient integration of diverse data sources to handle the complexity of varying crop and environmental conditions. Need for scalable models that adapt to different regional farming practices and soil types. Challenges related to data reliability and the infrastructure needed for continuous data collection and analysis	Highlights the importance of using machine learning and data analytics to enhance precision farming, motivating the development of adaptable and efficient frameworks to support farmers with actionable insights.
04.	Musanase, C., Vodacek, A., Hanyurwimfura, D., Uwitonze, A., & Kabandana, I. (2023). Data-driven analysis and machine learning-based crop and fertilizer recommendation system for revolutionizing farming practices. Agriculture, 13(11), 2141.	Data-Driven Analysis and Machine Learning-Based Crop and Fertilizer Recommendation System for Revolutionizing Farming Practices	This paper presents a comprehensive system that uses **data-driven machine learning algorithms** to recommend optimal crops and fertilizers. It integrates environmental and soil data, analyzing them to improve farm management. The approach enhances decision-making, leading to increased agricultural productivity and efficiency. The system's performance is validated through practical case studies	The need for more robust and adaptable models that can generalize across diverse environmental conditions. Challenges in data integration and handling variability in real-world scenarios. Limited focus on scalability and deployment in resource-constrained environments.	This study reinforces the value of using ML for precision farming, aligning with our project's aim to harness data analytics for optimizing agricultural practices and ensuring resource efficiency

[Table 1.2: Literature Survey]

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
05.	Akhter, R., & Sofi, S. A. (2022). Precision agriculture using IoT data analytics and machine learning. *Journal of King Saud University-Computer and Information Sciences*, *34*(8), 5602-5618.	Precision agriculture using IoT data analytics and machine learning	This paper discusses how various **IoT devices**, like sensors and devices(e.g., drones, soil moisture sensors, weather stations) are made the use of to gather real-time data, enabling farmers to make informed decisions. This collected data is then **processed**, using methods like *filtering and cleansing*, to ensure reliability and accuracy for further analysis. The authors discuss how machine learning algorithms enable **predictive analytics** for forecasting crop yields, diseases, and pests, while **decision support** systems help optimize irrigation, fertilizer use, and overall farm management.	*Data Management*: Challenges in storage, processing, and analysis necessitate robust data management frameworks. *Interoperability Issues*: Compatibility concerns among diverse IoT devices hinder effective implementation. *Cost Barriers*: High initial investment and maintenance costs can limit access for some farmers.	The paper is relevant today as it addresses the need for sustainable agricultural practices through the integration of IoT and machine learning technologies, enhancing productivity and resource efficiency
06.	Rajamanickam, J. (2021). Predictive model construction for prediction of soil fertility using decision tree machine learning algorithm. *INFOCOMP Journal of Computer Science*, *20*(1).	Predictive Model Construction for Prediction of Soil Fertility using Decision Tree Machine Learning Algorithm	This paper highlights the lack of knowledge among farmers about soil nutrients and methods for selecting suitable crops. Hence, this paper makes the use of supervised machine learning algorithms like **Decision Tree, K-Nearest Neigbour(KNN), and Support Vector Machine(SVM),** to predict soil fertility, focusing on macro- and micro- nutrient data. It concludes that the Decision Tree algorithm achieves highest accuracy of 99%, with minimum mean square error	The study mainly focusses on *nutrient data* and it neglects other factors like pest management, local agricultural practices, etc. It *lacks real time monitoring methods* for continuous analyzations of soil health, IoT devices could be introduced.	This paper provides an insight into machine learning applications that can enhance soil fertility, resulting in higher yields of crops.

[Table 1.3: Literature Survey]

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
07.	Trontelj ml, J., & Chambers, O. (2021). Machine learning strategy for soil nutrients prediction using spectroscopic method. *Sensors*, *21*(12), 4208.	Machine Learning Strategy for Soil Nutrients Prediction Using Spectroscopic Method	<u>ML Models for Soil Prediction</u>: The paper explores how machine learning models, particularly Principal Component Analysis (PCA), improve the accuracy of soil nutrient predictions. <u>Local Datasets for Better Accuracy</u>: It demonstrates that using local datasets (specific to a region) leads to higher prediction accuracy compared to general datasets. <u>Need for Region-Specific Models</u>: The study emphasizes the importance of developing models tailored to specific regions, as soil properties vary significantly across different areas, affecting prediction effectiveness	Dataset Limitation: The study uses a single dataset; incorporating more diverse datasets could improve model accuracy and generalizability. Missing Real-Time Data: The research does not integrate real-time environmental or weather data, which could enhance prediction accuracy.	This paper motivates the use of machine learning to improve soil fertility prediction, offering data-driven solutions to enhance agricultural productivity and sustainability.
08.	Talaviya, T., Shah, D., Patel, N., Yagnik, H., & Shah, M. (2020). Implementation of artificial intelligence in agriculture for optimisation of irrigation and application of pesticides and herbicides. *Artificial Intelligence in Agriculture*, *4*, 58-73.	Implementation of artificial intelligence in agriculture for optimisation of irrigation and application of pesticides and herbicides	Reviewed **AI-driven** irrigation and weeding techniques using drones and robots. These technologies addressed challenges posed by changing weather and rising food demand. AI technologies like image recognition and machine learning algorithms are applied to identify pest infestations and weed growth accurately	1. High initial investment and technology adoption barriers for small-scale farmers. 2. Need for reliable data collection and robust infrastructure.	This paper highlights AI's potential in sustainable farming, motivating further research to address water scarcity, rising food demand, and environmental challenges

[Table 1.4: Literature Survey]

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
09.	Shadrin, D., Menshchikov, A., Somov, A., Bornemann, G., Hauslage, J., & Fedorov, M. (2019). Enabling precision agriculture through embedded sensing with artificial intelligence. IEEE Transactions on Instrumentation and Measurement, 69(7), 4103-4113.	Enabling Precision Agriculture Through Embedded Sensing with Artificial Intelligence	The research paper presents a low-power embedded AI system designed to autonomously monitor plant growth for 180 days on a single **lithium-ion battery**, without any manual intervention.	Li-ion batteries are *costly*, and they are *sensitive to extremely high temperatures* as it affects the lifespan of the battery	This research paper is relevant as it showcases the integration of low-power AI technology in plant growth monitoring, addressing energy efficiency and sustainability challenges in agriculture.
10.	Priya, P. K., & Yuvaraj, N. (2019, November). An IoT based gradient descent approach for precision crop suggestion using MLP. In *journal of physics: conference series* (Vol. 1362, No. 1, p. 012038). IOP Publishing	An IoT Based Gradient Descent Approach for Precision Crop Suggestion using MLP	Developed a precision agriculture framework integrating IoT devices and a Multi-Layer Perceptron (MLP) model optimized using **gradient descent algorithm (Deep Learning algorithm)**. The framework collects real-time environmental and soil data through IoT sensors, which is then processed to provide crop suggestions tailored to soil and climatic conditions. The approach aims to enhance decision-making in farming, promoting better crop selection and boosting agricultural yield	-Limited scalability to broader agricultural applications. -The need for improvements in the efficiency and accuracy of data collection and processing methods. -Addressing challenges like the high initial cost and connectivity issues in remote areas.	Highlights using IoT and ML for data-driven precision farming, which aligns with our goal to enhance crop productivity and efficiency through smart, sustainable practices

[Table 1.5: Literature Survey]

SL. No.	Author Name, Journal Name, Vol., Page, Year	Title Of the Paper	Inference	Research Gap	Relevance
11.	Rekha, P., Rangan, V. P., Ramesh, M. V., & Nibi, K. V. (2017, October). High yield groundnut agronomy: An IoT based precision farming framework. In *2017 IEEE Global Humanitarian Technology Conference (GHTC)* (pp. 1-5). IEEE.	High yield groundnut agronomy: An IoT based precision farming framework	This study developed an IoT-based decision support system to optimize **groundnut farming**, offering personalized advice in farmers' regional languages through an **Android app**. By integrating real-time monitoring with accessible recommendations, it empowered farmers with practical, localized solutions to enhance yield and resource efficiency	The need for robust multilingual support for broader accessibility Issues like connectivity in remote areas and the high initial cost of IoT devices require further investigation	highlights the potential of IoT in precision farming, aligning with current efforts to develop practical, localized, and accessible solutions for farmers. Motivates further research to enhance agricultural productivity through smart technologies while addressing resource efficiency and farmer empowerment.

[Table 1.6: Literature Survey]

3. Problem Definition:

The absence in lack of implementation of technology in agro-based sector in a developing country like India where agriculture is considered as the backbone of the country's economy, a primary source of livelihood for a large portion of population, significantly contributing to the GDP and providing food security to the nation.
But the main problem arises when this age-old agricultural practice still continues in the fast-moving world, and completely relies on heavy labor, traditional planting and harvesting methods many a times leading to unnecessary utilization or overuse of resources with limited scalability in resource constrained areas.

4. Solution Strategy:

This problem can be encountered by usage of Precision Farming, a modern aided farming technique which leverages advanced technologies to increase productivity and efficiency in agriculture, reducing the wastage of resources by boosting their optimal use in an effective manner. This includes prediction of the right crop to be planted in respective season, gathers real time information about soil conditions, weather patterns, crop and more which is leveraged by a range of technologies such as sensors, GPS, artificial intelligence and data analytics which includes:

a. **Efficient Data Management**: Implement integrated, cloud-based data platforms that handle multi-source data for seamless analysis and scalability [2][6][10].
b. **Adaptable Models**: Develop flexible ML models that generalize well across different regions using techniques like transfer learning [3][4][7].
c. **Affordable Sensors**: Focus on creating low-cost, energy-efficient sensors and using community networks to minimize cost barriers [1][5][11].
d. **Improved Connectivity**: Utilize mesh networks, satellite technology, and edge computing for better connectivity in remote areas [8][9][11].
e. **Farmer Training**: Design user-friendly digital tools and provide training programs to simplify the adoption of precision agriculture methods [2][6][8].

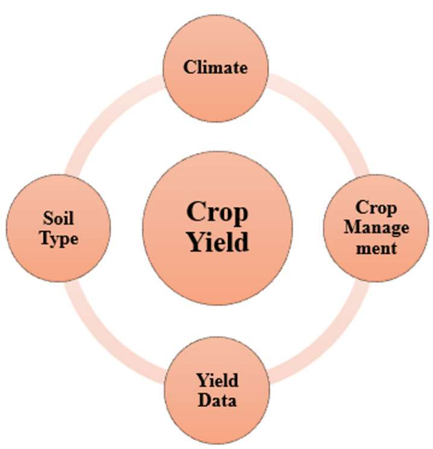
4.1. Findings:

The first and foremost step is to analyze the datasets, their features influencing crop growth such as soil moisture, weather variability, planting datasets, pest infestations and then normalize them accordingly, for input into various machine learning models for deployment.

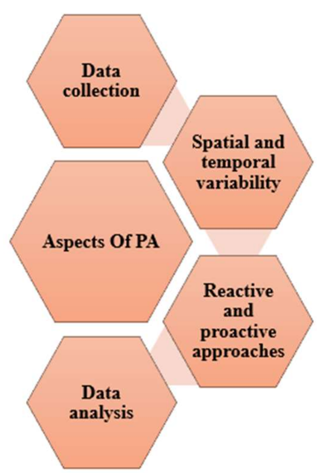
To get a systematic data set these methods should be followed so that no important features is missed out.

- Collaborate with agricultural research institutions, satellite data providers, and IoT companies to gather diverse datasets, including:
- Weather data (temperature, rainfall, humidity).
- Soil health parameters (pH, moisture, nutrient content)
- Remote sensing and satellite imagery for crop health monitoring. -
- Pest and disease outbreak records.

After fetching these datasets from various sources through various data integration techniques, removal of various outliers and noises should be done to get a clean and proper data set.



[Fig 3. Factors Affecting Crop Yield]



[Fig 4. Aspects of Precision Agriculture]

4.1.1. Datasets:

The following datasets are gathered from various sources to get a basic idea about the various crop patterns and their associated features: -

Sl. No.	Dataset Name	Dataset Author	Access Type and Link	Dataset Description
[i]	Soil Moisture Content of Sikkim District Wise [14]	Open Data by Govt. of India (data.gov.in)	[Open access] https://indiawris.gov.in/wris/#/soilMoisture	Moisture content of the soil across Sikkim
[ii]	Daily Rainfall Data (State wise and District wise) [15]	Indian Meteorological Department (https://mausam.imd.gov.in/)	[Open access] https://www.data.gov.in/resource/daily-data-imd-rainfall	Average rainfall in millimetre (MM).
[iii]	District wise, Season wise crop Production Statistics of India [16]	Indian Council of Agricultural Research(https://icar.org.in/)	[Open Access] https://www.data.gov.in/catalog/district-wise-season-wise-crop-production-statistics-0	Crop Production Details Season wise
[iv]	Crops water and Irrigation Water Requirements [17]	Indian Agricultural Research Institute (https://www.iari.res.in/en/index.php) Indian Council of Agricultural Research (https://icar.org.in/)	[Open Access] https://www.data.gov.in/resource/crop-water-and-irrigation-water-requirement-under-surface-and-drip-irrigation-methods	Contains Crop wise water irrigation requirements
[v]	Crop Recommendation Dataset [18]	Indian Chamber of Food and Agriculture(https://www.icfa.org.in/)	[Open access] https://www.kaggle.com/code/theeyeschico/crop-analysis-and-prediction/input	This dataset was built by augmenting datasets of rainfall, climate and fertilizer data available for India. Gathered over the period by ICFA, India.

[Table 2: List of Datasets Used]

[i] Soil Moisture Content of Sikkim District Wise:

- **Description:** This dataset describes about the soil moisture content, percentage and volume at 15cm across the state of Sikkim, district wise.
- **Features:** State Name, District Name, Avg. Soil Moisture level, Volume, Aggregate Soil moisture Percentage and Volume Soil Moisture Percentage

Showing 10 out of 10,700 records

Date	State Name	Districtname	Average Soilmoisture Level At 15cm	Average Soilmoisture Volume At 15cm	Aggregate Soilmoisture Percentage At 15cm	Volume Soilmoisture Percentage At 15cm
2018/06/30	SIKKIM	WEST	1.48	-	9.6	25.86159495
2018/06/30	SIKKIM	EAST	4.23	-	24.92	27.91799154
2018/06/30	SIKKIM	NORTH	2.04	-	13.94	22.19028228
2018/06/30	SIKKIM	SOUTH	2.12	-	13.84	27.32070217
2018/06/29	SIKKIM	WEST	1.55	-	10.09	26.2334694
2018/06/29	SIKKIM	EAST	4.43	-	26.12	28.56087538
2018/06/29	SIKKIM	NORTH	2.02	-	13.78	22.12144615
2018/06/29	SIKKIM	SOUTH	2.22	-	14.49	27.8273391
2018/06/28	SIKKIM	WEST	1.62	-	10.49	26.62717612

[Fig 5. Sample Data- Soil Moisture]

[ii] Daily Rainfall Data: (State wise and District wise)

- **Description:** Contains daily average data of all the states of India, district wise, giving an overview of the rainfall pattern of desired region.
- **Features:** State, District, Date, Year, Month, Average Rainfall

	State	District	Date	Year	Month	Avg_rainfall
0	Sikkim	East	01-01-2022	2022	1	0.000000
1	Sikkim	East	02-01-2022	2022	1	0.000000
2	Sikkim	East	03-01-2022	2022	1	0.000000
3	Sikkim	East	04-01-2022	2022	1	0.000000
4	Sikkim	East	05-01-2022	2022	1	0.000000
5	Sikkim	East	06-01-2022	2022	1	0.200484
6	Sikkim	East	07-01-2022	2022	1	0.002486
7	Sikkim	East	08-01-2022	2022	1	0.619652
8	Sikkim	East	09-01-2022	2022	1	0.428910
9	Sikkim	East	10-01-2022	2022	1	0.044614

Fig 6. Sample Data – Avg Rainfall In Sikkim

	Year	Month	Avg_rainfall
count	120.0	120.0	120.000000
mean	2022.0	1.0	1.281158
std	0.0	0.0	2.144392
min	2022.0	1.0	0.000000
25%	2022.0	1.0	0.000000
50%	2022.0	1.0	0.104285
75%	2022.0	1.0	2.065495
max	2022.0	1.0	11.745894

Fig 7. Sample Data – Description Of Avg_rainfall

	Year	Month	Avg_rainfall
count	5640.0	5640.0	5640.000000
mean	2022.0	1.0	1.109077
std	0.0	0.0	4.542948
min	2022.0	1.0	0.000000
25%	2022.0	1.0	0.000000
50%	2022.0	1.0	0.000000
75%	2022.0	1.0	0.000000
max	2022.0	1.0	61.164457

Fig 8. Sample Data – Description of avg rainfall throughout India

[iii] District wise, Season wise crop Production Statistics of India:

- **Description:** Contains detailed information about Season wise crop Production Statistics of India from year 1998 to 2015.
- **Features:** State, District, Crop Year, Crop, Area and Production

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
0	Sikkim	EAST DISTRICT	1998	Kharif	Maize	9690	16450
1	Sikkim	EAST DISTRICT	1998	Kharif	Oilseeds total	20	10
2	Sikkim	EAST DISTRICT	1998	Kharif	Rice	6690	9290
3	Sikkim	EAST DISTRICT	1998	Kharif	Small millets	1600	1520
4	Sikkim	EAST DISTRICT	1998	Kharif	Soyabean	1000	810

	State_Name	District_Name	Crop_Year	Season	Crop	Area	Production
709	Sikkim	WEST DISTRICT	2015	Kharif	Soyabean	546	520
710	Sikkim	WEST DISTRICT	2015	Kharif	Urad	914	811
711	Sikkim	WEST DISTRICT	2015	Rabi	Barley	12	11
712	Sikkim	WEST DISTRICT	2015	Rabi	Rapeseed &Mustard	625	540
713	Sikkim	WEST DISTRICT	2015	Rabi	Wheat	20	21

Fig 9. Sample Data of Various Crops Planted across Sikkim

[iv] Crops water and Irrigation Water Requirements:

- **Description:** This data set tells us about various crops and their water requirements through irrigation and rainfall
- **Features:** Crop and Irrigation Method, Total Crop Water Req. Irrigation Effective Rainfall (mm), Irrigation Water Req. (mm)

Showing 10 out of 17 records

Crop And Irrigation Method	Total Crop Water Requirement Irrigation Effective Rainfall Mm	Irrigation Water Requirement Mm
Maize (Kharif)- Surface Irrigation	550-600	300-350
Maize (Kharif)- Drip Irrigation	450-470	240-260
Wheat - Surface Irrigation	350-400	300-350
Wheat- Drip Irrigation	275-325	225-275
Rice (puddled)- Surface Irrigation	2300-2600	2000-2300
Rice (puddled)- Drip Irrigation	1100-1300	1400-1600
Ground nut (Summer)- Drip Irrigation	380-450	300-400
Ground nut (Kharif)- Drip	310-325	100-150
Mustard- Surface Irrigation	175-225	125-150
Moong Bean(Summer)- Drip	230-280	200-250

Fig 10. Sample Data of water required for various crops

[v] Crop Recommendation Dataset:

- **Description:** This dataset was built by augmenting datasets of rainfall, climate and fertilizer data available for India. Gathered over the period by ICFA, India.
- **Features:**
  - N - ratio of Nitrogen content in soil
  - P - ratio of Phosphorous content in soil
  - K - ratio of Potassium content in soil
  - temperature - temperature in degree Celsius
  - humidity - relative humidity in %
  - pH - pH value of the soil
  - rainfall - rainfall in mm

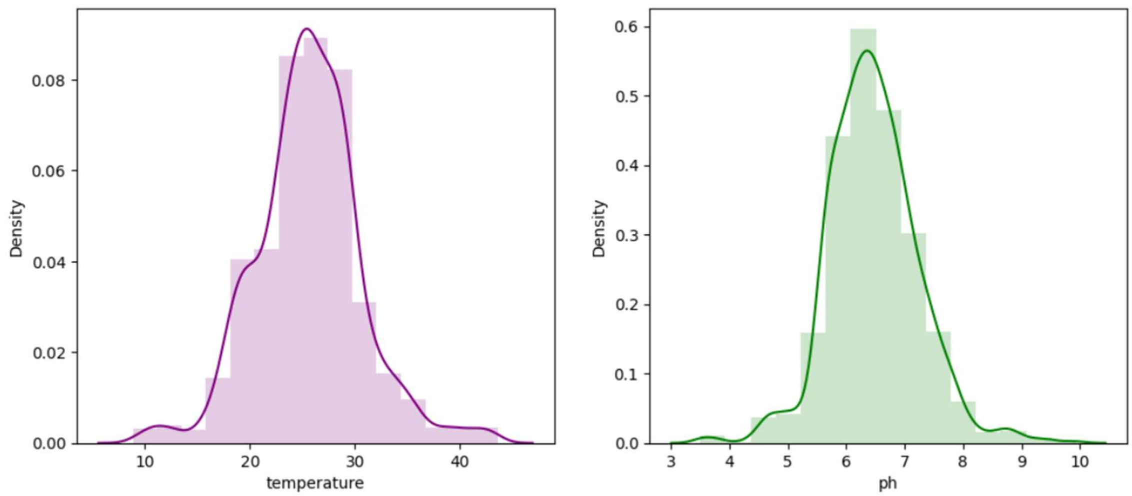
	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice
5	69	37	42	23.058049	83.370118	7.073454	251.055000	rice
6	69	55	38	22.708838	82.639414	5.700806	271.324860	rice
7	94	53	40	20.277744	82.894086	5.718627	241.974195	rice
8	89	54	38	24.515881	83.535216	6.685346	230.446236	rice

Fig 11. Sample Data of various crops and their needed parameters

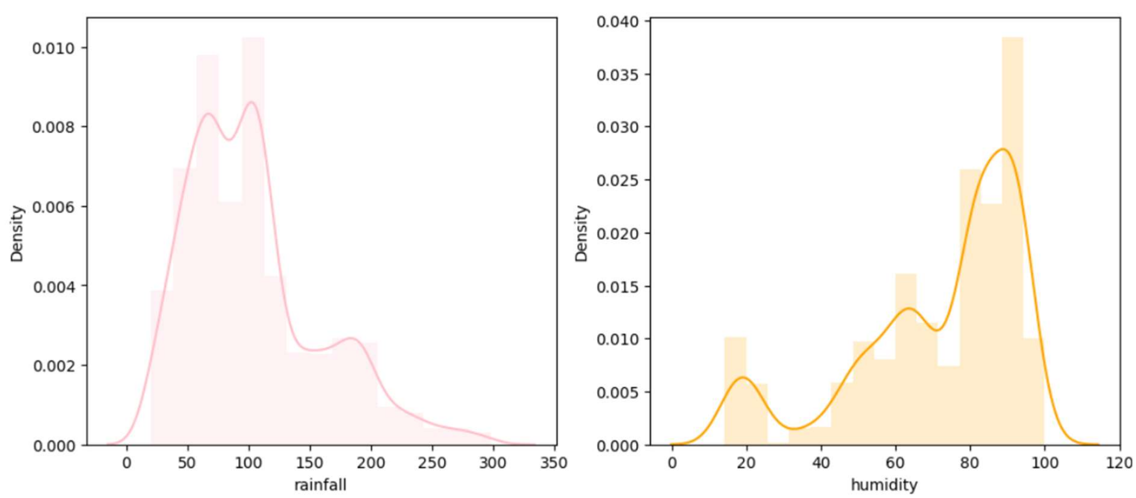
#Visualization Of Crop Recommendation Dataset:

- An overview of effect of various parameters such as ph, temperature, humidity and rainfall on the crops/labels. (Here, histograms are plotted to check if the data set is balanced or not, if not the dataset should be down sampled)

<Axes: xlabel='ph', ylabel='Density'>



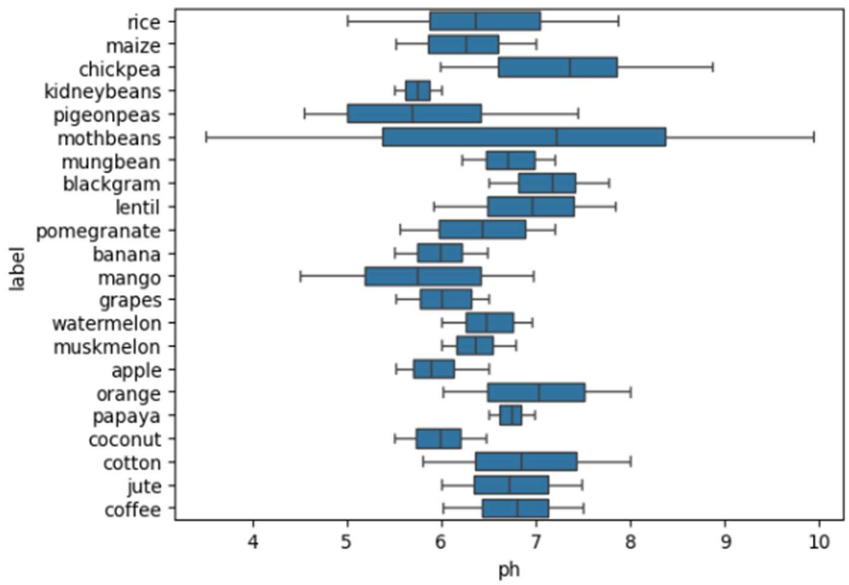
<Axes: xlabel='humidity', ylabel='Density'>



[Fig.12. Vizualization of Various Parameters]

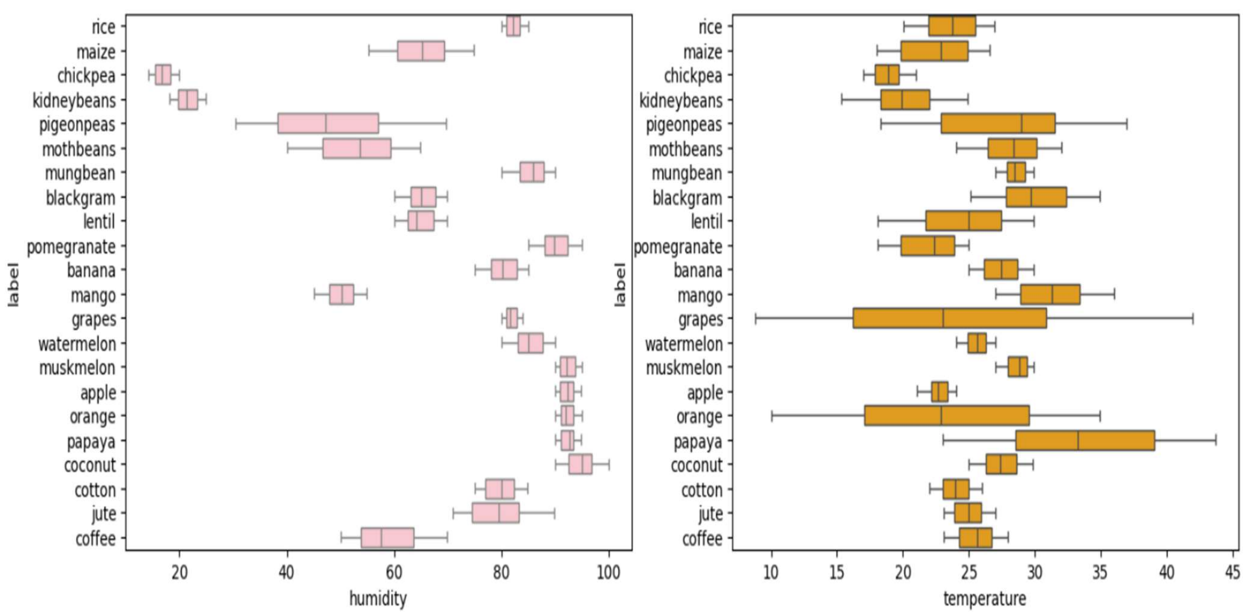
- A better idea about the ranges of these parameters can be obtained using boxplot:

```
<Axes: xlabel='ph', ylabel='label'>
```



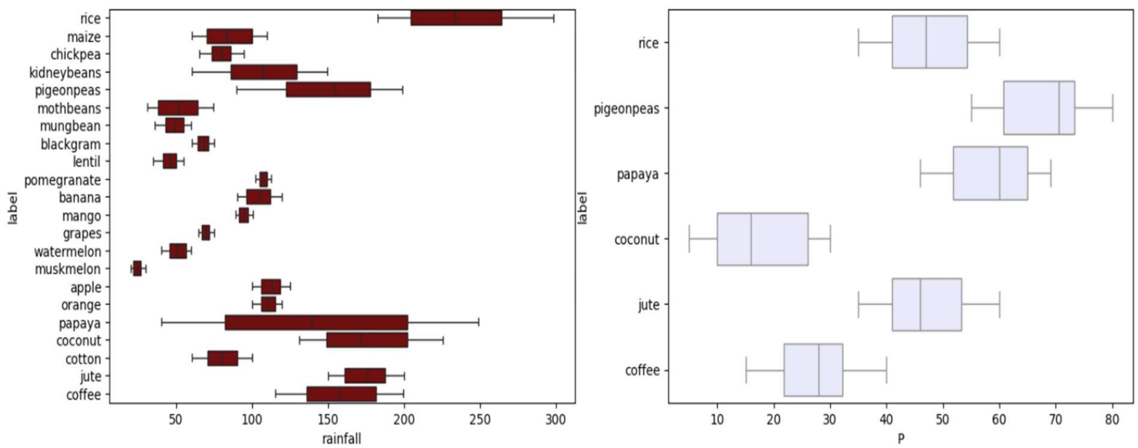
[Fig.13. Boxplot of the labels vs. pH]

Inference: pH values are critical when it comes to soil, A stability between 6-7 is preferred.



[Fig.14. Boxplot of the labels vs. humidity & temperature]

Inference: Most of the crops are prone to humid climate except for crops like chickpeas, beans, and other grains because they basically grow in arid climate. The mode range for humidity lies between 70-80 mm. While most of the crops grow in moderate temperature i.e, around 20-30 deg. centigrade; few crops such as papaya grows in a very hot and humid climate.



[Fig.15. Boxplot of the labels vs. rainfall and its impact on Phosphorus(P)]

Inference: The above subplot gives us an idea about the rainfall patterns followed by these variety of plants over the years, and an interesting impact on Phosphorus is studied that Phosphorous levels are quite differentiable when it rains heavily (above 150 mm) for the crops of rice, coconut, pigeon peas, papaya, jute and coffee.

5. Objective:

The key outcomes to be expected from this project would be to utilizing data-driven machine learning techniques for more precision farming. Optimize resource allocation based on an analysis of data, both historical and real-time, to ensure efficient distribution of water, fertilizers, and pesticides, thereby avoiding waste and maximizing yields of crops. This includes predictive analytics models for predicting crop health and growth patterns, and potential disease outbreaks, so the farmer can make proper choices about planting schedules, crop rotation, and pest management strategies.

Additionally, continuous monitoring techniques based on IoT and spectroscopic methods are used to monitor soil and crops for optimal crop productivity through actionable insights. The project, therefore, aims to study the economic viability of precision farming systems through data-driven assessments, to ensure stakeholders could opt for cheap strategies to reduce costs and increase yields.

6. Feasibility and Viability:

Modern technological advancement well supports the feasibility and viability of the proposed precision farming system. IoT devices, sensors, satellite data along with the integration of machine learning algorithms have now been well demonstrated in agriculture as well as in other sectors, utilizing cloud computing and AI platforms and permitting real-time monitoring and predictive analytics. Therefore, the system becomes technologically feasible. The system further is highly scalable and, therefore, scales across multiple types of crops, geographic regions, and climatic conditions that it may be required to function within, which strengthens the utility across different types of farming practices. Further, the use of customizable machine learning models trained upon region-specific datasets drives broader applicability and the capacity to effectively meet local agricultural requirements.

7. Challenges And Bottlenecks:

- Access to relevant and well-structured datasets is one major problem, especially in rural and developing regions. Scarcity of information about soil health, weather, and crop yield during all seasons has a negative impact on data-driven precision agriculture. On large-scale and high-quality datasets generation, collection, and sustaining are conventionally short of support for infrastructure and funding.

- A lack of technical knowledge poses serious challenges in adoption and use of progressive systems. They are not fully aware of analyzing data and working on machine learning models. This has substantially hindered fast integration and utilization of modern technology for increasing agricultural productivity.

- Missing internet connectivity, unstable power supply, and no direct access to digital infrastructure in rural areas greatly inhibit the rollout of smart farming solutions. That further makes it difficult on the usability of analytics tools that usually require real-time data transmission, analysis, and monitoring.

- Extraction and sharing of agricultural data raise questions on their privacy, ownership, and security. Farmers would hardly know who owns their data and how it may be used beyond certain limits they have drawn. It is without any specific frameworks that trust in data-driven systems may become eroded.

- The areas dominated by small and fragmented farms present economic and scalability challenges. Because precision farming technologies often require a certain scale of operation to enable viable financial returns, smallholders will have a difficult time in adopting them freely without ongoing outside support or cooperation.

7.1. Strategies to Overcome These Challenges:

1. Data Privacy and Ownership
   It is important to raise a secure and appropriate system through which possibly intrusive data collections and sharing could allow for farmers' data protection. The farmers-opaque data policy should really be a specific focus as this may be eroded when there are third parties in play. Educating the farmers on their data rights and providing them with tools to manage their data will further fortify the trust.

2. Small and Fragmented Landholdings
   Models like cooperative farming provide consolidation of smaller plots, empower farmers to share resources and technologies, allow for scaling, and make precision farming economical. Government incentives or subsidies may encourage farmers to participate in such partnerships/relationships.

3. Building Data Infrastructure
   Governments, research institutions, and agricultural technology companies should act in coalition to aggregate soil conditions, climate, and crop performance data in a centralized repository so that stakeholders may access this information to make decisions and train models.

4. Training and Education Programs
   Training programs must be tailor-made for farmers so that they learn and use data-driven tools effectively. Availability of workshops, hands-on sessions, and e-learning platforms in the local language would ease the process of introducing new technology. User-friendly software development would ensure accessibility even for people without much technical expertise.

5. Investing in Rural Infrastructure
   Improving connectivity and reliable power supply in rural areas would have a significant impact on real-time data collection and analysis. Investment in rural infrastructure would facilitate a much swifter adoption of precision farming tools to the extent that farmers can access these technologies uninterrupted.

6. Data Privacy Regulations
   Establishment of concrete legal frameworks governing data privacy and security. The government and policymakers must enshrine these regulations into law in a manner that protects farmers' rights, advocates ethical data usage, and ultimately helps build trust within the whole data-sharing initiative.

7. Cooperative Farming Models
   Drilling down to cooperative farming models will aim to deal with inefficiencies occasioned by fragmented landholdings. Farmers, pooling together their limited resources and efforts, can access modern equipment, lower their respective costs, and boost productivity jointly.

8. Impacts And Benefits:

1. Enabling high agricultural productivity and enhanced profitability
   The synergetic outcome of enhanced crop yields and improved resource management translates into higher profit margins for farmers. The precision agriculture approaches often save farm operation costs through more efficient input-output relationships.

2. Environmental protection and sustainability
   A reduction in agrochemical and water usage minimizes environmental impacts for sustainability. Such approaches ensure ongoing health of ecosystems and farmland to promote agricultural productivity into the future.

3. Informed decision-making
   Adding data into any decision-making process improves such processes and allows for inline decisions for the farmer. Farmers can do better risk management ranging from early pest infestations or diseases or nutrient deficiencies which that will lead to ensuring crops for maximum profit.

4. Climate resilience
   In view of the advances in technology, it has been suggested that precision agriculture techniques reduce the vulnerability of crops to extreme climatic disturbances and enhance agricultural production and processes.

5. Economic and operational efficiencies
   Precision agriculture provides cost-effective interventions for farms of any scale. The customized and scalable tools enable the small and large producers to access advanced farming practices; hence, a source of income in rural areas.

6. Overall benefits to both farmers and society
   The enables blended effect of increased profitability sustainability and resilience overall will make the contributing door open for improved livelihoods for the farming community as food security and resource conservation for society at large.

9. Action Plan:

I. Data Collection and integration:

The first step involves bringing together stakeholders including agricultural research institutions, satellite data providers, and IoT firms in order to acquire diverse datasets crucial for precision farming. These datasets consist of:

a) Weather Data: Analysed data such as temperature, rainfall, and humidity so as to determine environmental settings influencing crop growth.
b) Soil Health Parameter: Parameters such as pH, soil moisture, and fertilizing conditions combined to ascertain and improve the quality of soil, therein achieving optimization in agriculture activities.
c) Remote Sensing and Satellite Imagery: Used in high-resolution images to monitor crop health, while zoning in on stressed areas and development patterns over time.
d) Pest and Disease Outbreak: Historical records of pest and disease outbreaks. This will aid in predictive modelling of the likely occurrence of future incidences.

II. Feature Engineering:

This step involves extracting features that are highly influential in determining the performance of crops to develop input spaces upon which the machine-learning model will depend.

a) This includes evaluating soil moisture, weather variability, planting schedules, and pest infestation and how each may affect crop growth.
b) Data preprocessing procedures such as normalization and cleaning act as a prerequisite for the effective analysis and functional prediction of the raw data.

III. Selection of Algorithm:

Selecting an algorithm is one of the most critical steps toward the solution of varied challenges in precision farming. Some methods considered include:

a) **K-Nearest Neighbours-k-NN**, for classification and regression problems such as grouping crops based on their similarity or predicting crop yield.
b) **Support Vector Machines-SVM**, for complex classifying problems like soil classification or pest occurrence.
c) **Decision Trees**, for interpretability and branching decisions that assist in determining the most favourable planting or irrigation strategies.
d) **ARIMA, Auto-Regressive Integrated Moving Average**: A statistical model for time-series forecasting, like predicting the distributions of rainfall patterns or season yields.
e) **Gradient Descent**: An optimization technique to speed up the training process by which a model minimizes errors in prediction.

10. Gantt Chart:

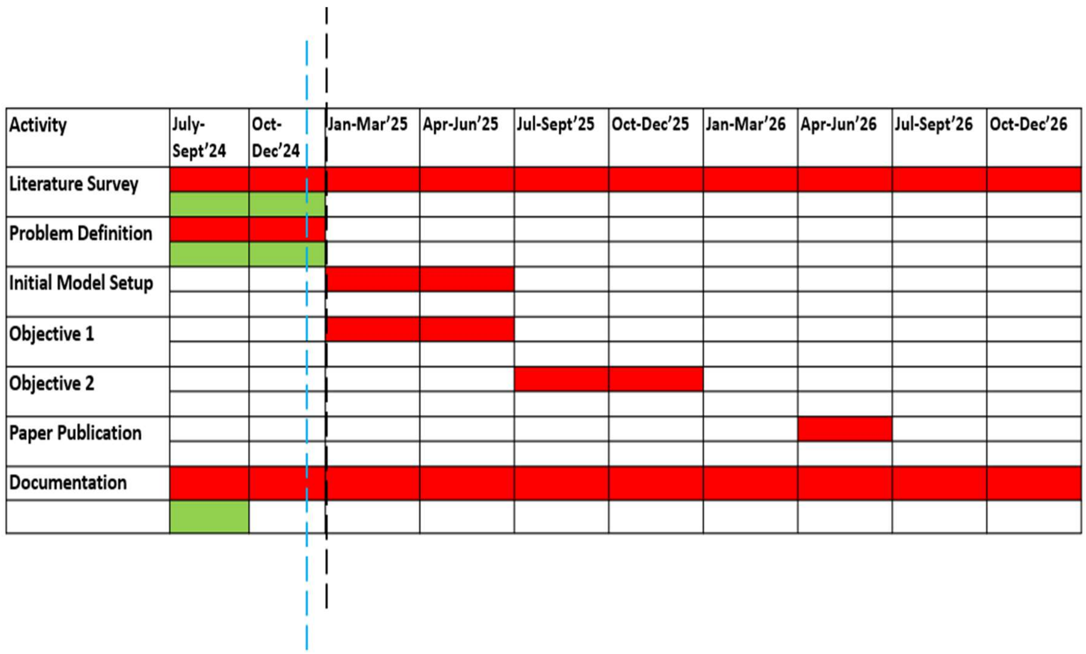


Fig 6 : Gantt Chart

- - - - : Ongoing Period   : Proposed Activity   : Completed Activity   - - - - : Completed Period

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